

A Brain to UAV Communication Model using Stacked Ensemble CSP algorithm based on Motor Imagery EEG signal

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Abstract—The creation of non-invasive Neuro-headset technology has resulted in a variety of applications such as control, automation, and vehicular control. Unmanned Aerial Vehicles (UAVs), mostly used in aerial communication, is a growing field of study that may be applied to both defense and commercial applications. As the need for drone control grows, recent advancements in the BCI-based drone control system have been made. This paper proposes a unique prototype to achieve smooth and stable control of a UAV by the human brain based on a motor imagery signal for semi-autonomous navigation. In such cases the accuracy and signal detection and response time get the utmost priority to get a decent performance of the vehicle. Currently, it is very common to use CSP for the classification of motor imagery data. However, this algorithm needs improvement in the low quantity of training samples and noisy data. Because in that scenario the over-fitting may be a problem. To overcome those drawbacks this study proposed Stacked Ensemble Common Spectral Pattern (SECSP). Comparing this method with Common spatial pattern (CSP), Filter Bank CSP (FBCSP), and Common Spatio-Spectral Patterns (CSSP) it has been proved that our model outperforms those popular methods in terms of accuracy, sensitivity, robustness, and precision for motor imagery Electroencephalogram data. The publicly available data-set for motor imagery BCI Competition IV, Data-set 1, BCI Competition III, Data-set IVA have all been tested with Stacked Ensemble CSP (SECSP). Performance is enhanced when compared to previous approaches, with an average accuracy of 87.74 percent for all subjects and 95.64 percent for the first and second data sets, respectively.

Index Terms—UAV, Aerial Communication, EEG, SECSP, BCI, Motor Imagery

I. INTRODUCTION

For a long time, the world has been attempting to link the human brain and machines. Linking the human brain with the rest of the world is always been a unique strategy on which researchers have already started making different prototypes, it is useful in a variety of applications and has consequently acquired substantial traction in recent years [1]. This research [2], [3] started to get attention when the researchers realized that these can be an excellent and efficient way to help physically disabled people interact with the rest of the world. BCI technologies transform EEG, generated by control signals

of the brain allowing operation of external devices without any muscle involvements [4]. These early experiments established the framework for later research teams to create more complex brain-controlled UAV prototypes so that maximum accuracy can be achieved. Furthermore, BCI has created a breakthrough framework that enables the development of real-time robotic devices controlled by brain signals [5]. This covers assistance equipment for persons with impairments, like wheelchairs and prosthetic limbs, as well as various robotic systems designed to help healthy people daily [1]. Drone applications have grown in prominence in recent years as a result of their unusual intervenes in executing on-air executions when the access is done by any pilot is not feasible. The innovative notion of creating a brain-guided UAV sparked several continuing experiments spearheaded by experts at the University of Minnesota [6]–[8]. This research presents a BCI system for achieving continuous flying of an Unmanned Aerial Vehicle (UAV) in an interior 2D environment. In the research related to UAV, the HI (human interface) is an interface that allows the controller to generate commands or instructions and manage the state space of the aircraft. Human Machine Interface (HMI) acts like a control plane, such as a portable joystick, a touchscreen, or any standard hardware configuration for the interface, in its most basic form. BCI is a kind of sophisticated human-machine interface that has its advantages. Many researchers have lately needed to concentrate on hovering autonomous navigation in real space utilizing BCI, as it allows for flexible movement in 2D and even 3D space, and also with any distant devices with informative contact to the real world might serve as the replacement [9].

The motivation behind our work: UAVs are becoming increasingly important in today's environment [10]. Wireless applications have been more popular in the scientific, educational, and hobbyist communities in recent years. Environmental monitoring, such as data collection in distant places, may be done by unmanned aerial vehicles (UAVs). Aerial platforms have greater maneuverability and passability than wheeled platforms, and a swarm of robots might take

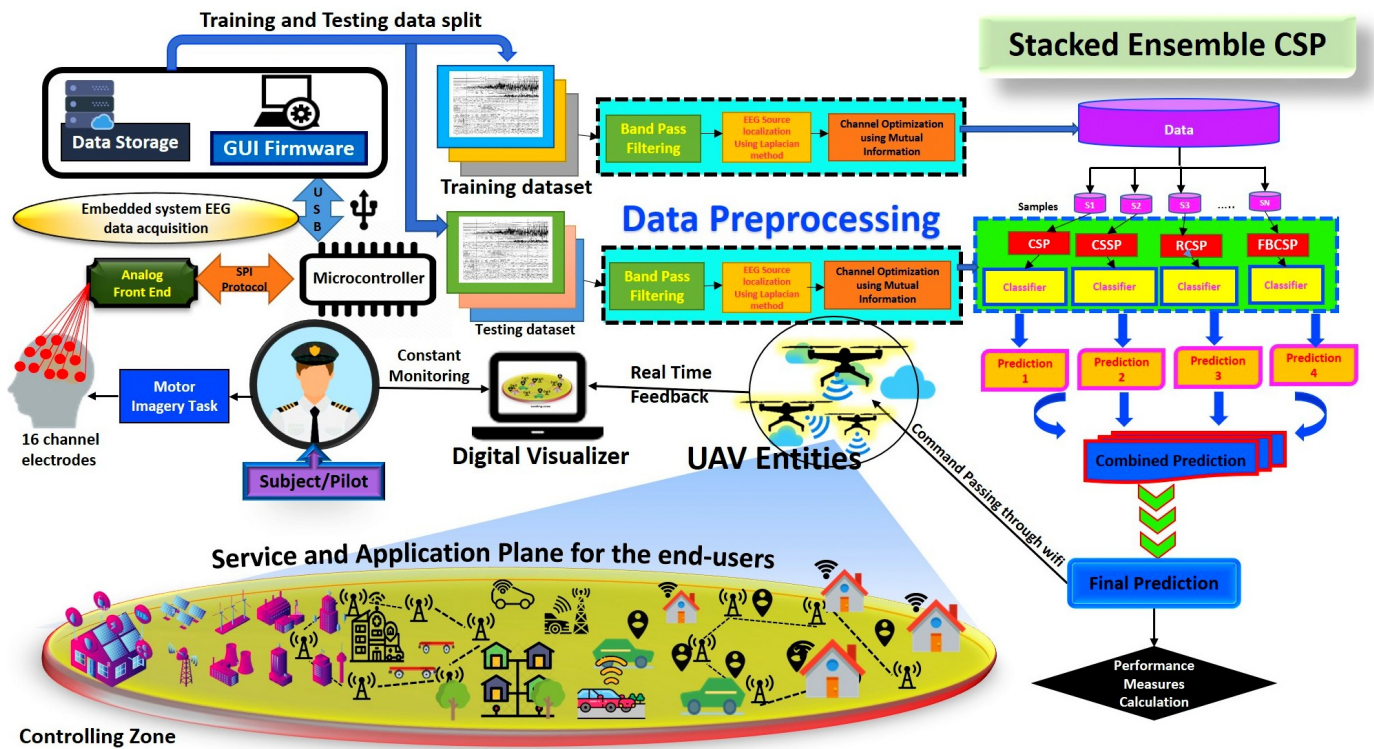


Fig. 1: SECSP based model architecture for classification motor imagery task

up all of the heavy liftings in certain conditions [11]. UAV control takes certain technical abilities, but skilled individuals can do so. Brain-controlled UAVs are a real challenge in today's date. As the research on brain signals gains popularity several methods for the brain-computer interface based on EEG signals have been incorporated to make a smart system [12]–[15]. By this process, a communication channel can be created by which the information can pass from brain to machine while no strong muscle action is present in the body of the subjects. The electrical signal generated in neurons is transformed into a command or message and then message is decoded by the machine or computer to understand the brain activity.

Novelty and Contribution: Our proposed method followed the Common Spatial Pattern. To date, various advancement has been done to make the method more reliable and accurate. Each updated version has overcome different cons of CSP. For example, to select the optimal temporal frequency band for CSP, common spatio-spectral patterns (CSSP) [16] approach was introduced which includes 1 time delayed sample with the previous algorithm to optimize a filter. They have worked on filters and instead of using a simple filter, they have used an arbitrary Finite Impulse Response Filter for optimizing simultaneously. Another type of CSP is the Spectral weighted CSP (SPEC-CSP) [17] algorithm which first optimizes the temporal filter and then optimizes the spatial filter alternatively. [18] proposed Filter bank Common Spatial Pattern to enhance the performance of the CSP method by applying the algorithm

of automated selection of frequency range which is subject-specific for band-pass filtering. But this method is only capable to classify two classes of motor imagery data.

II. METHODOLOGY

A. Dataset Used

For this experiment, we have downloaded two motor imagery EEG data sets which are publicly available online. The two data sets are part of the BCI competition and they are available as BCI Competition Data set 1 and BCI Competition III - Data set IVA.

B. EEG Data Pre-processing

1) *Butter-worth Band-pass Filtering:* For motor imagery classification we need to figure out which band of the EEG is required. From the previous studies, it has been noticed that when the subject involves muscle movement or imagines any muscle movement the μ and β band contain the required information. To extract any bands from EEG signal we need some kind of filtering. In our study, the data sets are segmented into single trials and it comes out that each segment lasts for 4 sec. To extract the μ and β band we have applied the Butterworth Band-pass filter with the order of 3. From those bands, we have got the motor imagery information more accurately.

2) *Source Localization using high laplacian Method:* One of the key focuses of tackling the inverse problem in EEG has been source localization. This entails determining where in the brain a specific sort of activity originates based on surface EEG recordings [19]. In neuroscientific analyses as

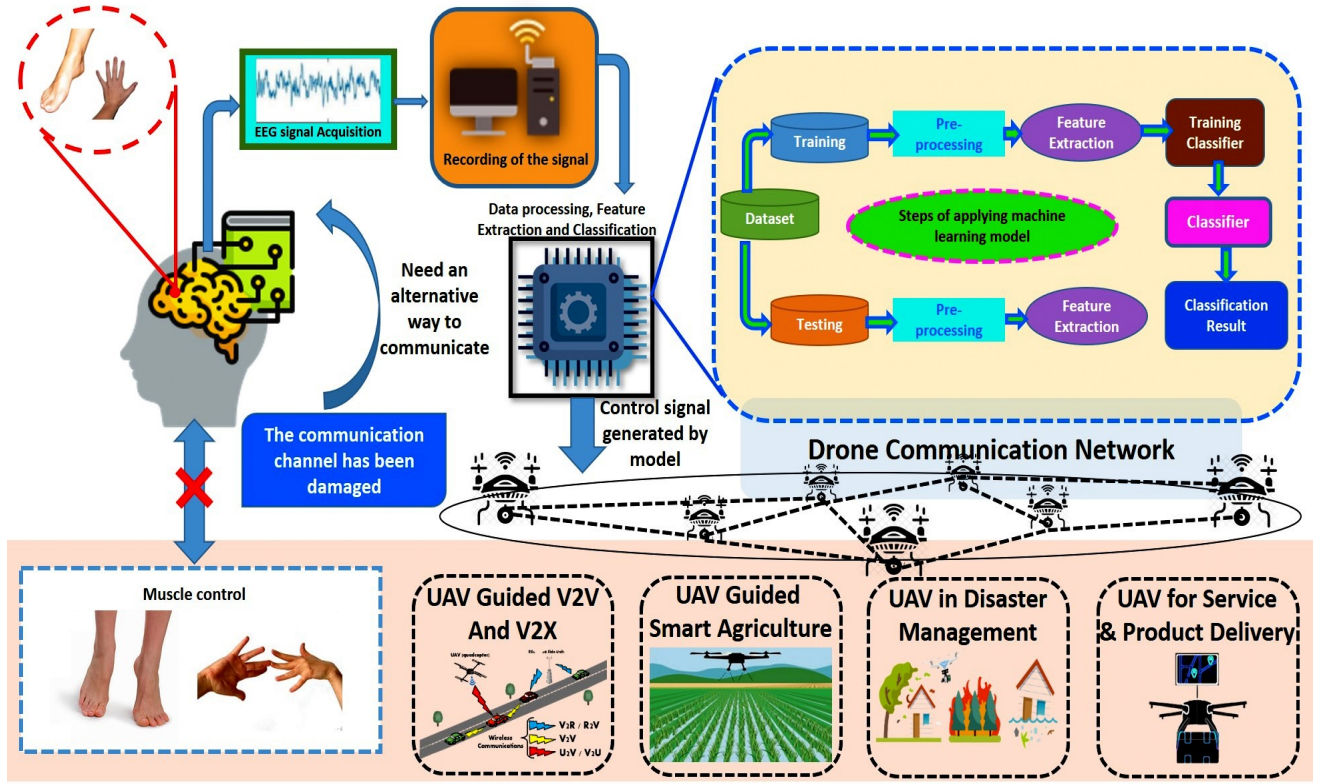


Fig. 2: An overview of the application process of the motor imagery signal to control various devices

well as numerous therapeutic applications, it is necessary to estimate the EEG source location with high precision [19]. In the case of UAV control, the model has to be accurate so it's a crucial step to do. In our experiment, we have used a High Laplacian Filter in the later step. There are various methods are available for source localization of EEG. For example CAR, Low and High laplacian. After testing with 3 popular methods it has been concluded that high laplacian is performing best with both of the datasets. The mathematical expression for the Laplacian filter [20] is as follows:

$$v_k^L(t) = v_k(t) - \sum_{m \in N_k} \partial_{km} v_m(t) \quad (1)$$

$$\partial_{km} = \frac{1/d_{km}}{\sum_{m \in N_k} 1/d_{km}} \quad (2)$$

3) *Channel Optimization using Mutual Information:* When we are trying to retrieve information from an EEG signal it may not be necessary to consider all of the channels which have been used to record the signal. Because most cases a particular lobe or zone of the brain is dedicated to a definite task. So some of the channels contain redundant information that may not be required for this experiment. We need to select those channels only which contains maximum information for motor imagery task. Also, EEG classification may be made quicker, more accurately, and more simply by using channel selection. Furthermore, channel selection

improves model generalization, reduces the dimensionality impact, and speeds up the learning process. In this study, we have selected the channels by mutual information theory. In this method, the channels are selected individually based on mutual information. But the main challenge lies in choosing the proper channels as it is mainly subject or person dependent. Distinct channels likely carry important information regarding the motor imagery challenge for each subject or person. The channels of the EEG which carries maximum information are considered. Wavelet Transform was applied to avoid the problem that happened due to its non-stationary nature. After the transformation the power of coefficients is extracted and depending on the maximum information contained by channels, the channel optimization has been done. The assessment of reciprocal dependency (amount of information) between two variables is known as mutual information (MI). The information entropy of random E can be defined as: Given a discrete random variable E, which contains $e_1, e_2, \dots, e_n$ random states with probability distributions of $p_e(e_1), p_e(e_2), \dots, p_e(e_n)$, the information entropy of random E can be defined as:

$$H(E) = - \sum_{i=1}^n p_e(e_i) \log_2 p_e(e_i) \quad (3)$$

Likewise, random F's information entropy may be written as:

$$H(F) = - \sum_{i=1}^n p_f(f_i) \log_2 p_f(f_i) \quad (4)$$

TABLE I: Comparison of accuracies with the RF classifier for the dataset BCI Competition III, Dataset IVA

Methods Applied	aa	al	av	aw	ay	Mean	SD
CSP+ Random Forest	78.52 %	92.35%	62.47%	92.15%	85.21%	82.14%	12.41%
CSSP+ Random Forest	78.25%	91.75%	65.47%	95.45%	87.3%	83.64%	12.33%
RCSP + Random Forest	79.23%	98.78%	66.21%	89.25%	83.65%	85.23%	12.54%
FBCSP + Random Forest	77.25%	95.45%	69.41%	82.48%	82.41%	83.25%	13.26%
Stacked Ensemble CSP+ Random Forest	87.23%	98.45%	68.65%	86.15%	84.26%	82.14%	12.96%
CSP+ Random Forest, boosting	77.44%	93.24%	64.2%	83.25%	76.26%	83.6%	11.55%
CSSP+ Random Forest, boosting	78.25%	94.33%	68.15%	92.19%	81.52%	80.25%	11.32%
RCSP+ Random Forest, boosting	82%	92.38%	69.35%	94.34%	88.21%	81.34%	12.9%
FBCSP+ Random Forest, boosting	78.11%	94.10%	68.15%	91.16%	84.32%	85.21%	11.35%
CSP+ Random Forest, bagging	77.24%	96.12%	65.12%	93.65%	86.32%	82.96%	11.25%
CSSP+ Random Forest, bagging	79.45%	96.47%	68.45%	92.11%	81.32%	83.25%	11.59%
RCSP+ Random Forest, bagging	82.45%	96.58%	69.55%	95.24%	88.41%	82.44%	12.69%
FBCSP+ Random Forest, bagging	78.45%	94.21%	67.25%	95.36%	85.63%	85.41%	12.35%
Stacked Ensemble CSP+ Random Forest, bagging (Proposed Method)	89.21%	96.42%	68.14%	95.3%	89.65%	87.74%	12.99%

TABLE II: Comparison of accuracies with the SVM classifier for the dataset BCI Competition III, Dataset IVA

Methods Applied	aa	al	av	aw	ay	Mean	SD
CSP+ SVM	71.55 %	95.35%	63.46%	93.13%	85.23%	86.40%	12.63%
CSSP+ SVM	76.28%	92.56%	62.43%	95.25%	82.55%	89.23%	12.32%
RCSP + SVM	72.29%	93.68%	63.25%	88.22%	82.56%	82.23%	13.14%
FBCSP + SVM	78.59%	95.12%	69.28%	82.68%	83.51%	83.11%	12.46%
Stacked Ensemble CSP+ SVM	86.93%	92.45%	68.63%	89.15%	84.26%	88.54%	12.36%
CSP+ SVM, boosting	73.26%	93.32%	66.22%	93.95%	86.22%	83.3%	11.25%
CSSP+ SVM, boosting	72.55%	91.47%	68.25%	91.29%	86.2%	85.56%	12.79%
RCSP+ SVM, boosting	78.55%	93.28%	63.26%	95.11%	82.34%	83.11%	12.92%
FBCSP+ SVM, boosting	79.11%	92.61%	62.25%	95.26%	82.33%	84.32%	11.15%
CSP+ SVM, bagging	72.4%	91.22%	61.12%	93.95%	86.22%	82.23%	13.75%
CSSP+ SVM, bagging	72.95%	96.47%	68.35%	92.69%	86.32%	85.55%	11.79%
RCSP+ SVM, bagging	78.35%	93.28%	63.56%	95.34%	81.34%	83.41%	12.19%
FBCSP+ SVM, bagging	79.25%	93.31%	63.25%	95.66%	83.73%	85.33%	13.15%
Stacked Ensemble CSP+ SVM, bagging (Proposed Method)	89.14%	96.42%	69.96%	95.65%	89.65%	84.74%	12.69%

TABLE III: Comparison of accuracies with the RF classifier for the dataset BCI Competition IV, Dataset I

Methods Applied	a	b	c	d	e	f	g	Mean	SD
CSP+ Random Forest	82.56 %	62.55%	70.61%	69.12%	88.32%	84.23%	83.26%	72.62%	12.03%
CSSP+ Random Forest	71.24%	62.3%	65.6%	67.85%	82.15%	87.36%	84.36%	73.12%	13.23%
RCSP + Random Forest	79.14%	74.33%	67.78%	64.23%	81.3%	81.36%	87.26%	78.2%	11.20%
FBCSP + Random Forest	77.36%	71.29%	69.52%	66.14%	82.63%	83.58%	84.69%	74.36%	11.41%
Stacked Ensemble CSP+ Random Forest	83.11%	76.63%	64.26%	67.25%	89.36%	89.22%	95.36%	80.45%	12.86%
CSP+ Random Forest, boosting	71.13%	78.75%	63.19%	66.7%	81.36%	81.63%	87.26%	81.16%	11.3%
CSSP+ Random Forest, boosting	76.63%	73.76%	67.88%	62.93%	91.32%	85.36%	86.65%	83.12%	9.76%
RCSP+ Random Forest, boosting	88.12%	77.46%	63.06%	61.2%	83.25%	82.14%	82.36%	81.64%	11.10%
FBCSP+ Random Forest, boosting	82.63%	79.36%	66.24%	62.55%	84.98%	86.21%	84.36%	89.13%	12.50%
CSP+ Random Forest, bagging	86.15%	79.62%	68.36%	61.95%	89.15%	84.23%	87.33%	87.35%	11.50%
CSSP+ Random Forest, bagging	76.59%	66.22%	77.63%	61.36%	86.47%	86.11%	81.96%	88.44%	9.82%
RCSP+ Random Forest, bagging	76.52%	89.98%	63.14%	64.86%	83.74%	82.36%	84.68%	83.16%	11.90%
FBCSP+ Random Forest, bagging	77.39%	76.25%	71.36%	62.99%	89.21%	85.36%	92.6%	84.76%	9.55%
Stacked Ensemble CSP+ Random Forest, bagging (Proposed Method)	89.19%	87.14%	78.76%	67.66%	90.36%	91.26%	98.36%	95.64%	11.56%

So the equation for joint entropy will be:

$$H(E, F) = - \sum_{i=1}^n p_{ef}(e_i f_i) \log_2 p_{ef}(e_i f_i) \quad (5)$$

Mutual information between two-channel signals can be expressed in the following way:

$$MI(E, F) = H(E) + H(F) - H(E, F) \quad (6)$$

TABLE IV: Comparison of accuracies with the SVM classifier for the dataset BCI Competition IV, Dataset I

Methods Applied	a	b	c	d	e	f	g	Mean	SD
CSP+ SVM	52.15 %	62.36%	60.41%	63.19%	87.11%	80.23%	84.16%	73.62%	14.23%
CSSP+ SVM	71.12%	79.31%	62.96%	64.75%	83.45%	78.29%	89.12%	77.62%	13.25%
RCSP + SVM	75.24%	73.28%	65.65%	68.21%	88.36%	82.55%	88.53%	72.36%	9.32%
FBCSP + SVM	79.29%	75.42%	63.35%	62.18%	83.34%	89.41%	73.26%	78.36%	11.46%
Stacked Ensemble CSP+ SVM	84.17%	72.5%	63.66%	62.59%	91%	91.44%	90.1%	80.25%	13.3%
CSP+ SVM, boosting	77.15%	77.25%	62.14%	65.78%	82.36%	82.38%	85.21%	79.26%	10.56%
CSSP+SVM, boosting	72.35%	71.26%	61.48%	64.33%	86.31%	90.32%	77.65%	77.25%	12.76%
RCSP+SVM, boosting	82.12%	77.14%	62.86%	67.32%	86.14%	89.65%	79.65%	80.36%	12.51%
FBCSP+SVM, boosting	82.11%	78.22%	61.24%	62.85%	83.45%	88.12%	82.15%	79.23%	11.52%
CSP+ SVM, bagging	84.45%	79.72%	67.62%	61.55%	85.32%	83.13%	83.26%	77.65%	12.75%
CSSP+ SVM, bagging	71.65%	76.42%	67.35%	61.9%	87.68%	85.56%	84.96%	82.36%	11.88%
RCSP+ SVM, bagging	73.59%	69.18%	63.64%	65.96%	82.68%	81.11%	84.33%	85.36%	12.39%
FBCSP+ SVM, bagging	72.26%	73.41%	61.65%	62.5%	81.4%	85.43%	91.46%	85.36%	11.55%
Stacked Ensemble CSP+ SVM, bagging (Proposed Method)	84.19%	82.14%	68.76%	65.66%	89.65%	83.04%	91.21%	89.21%	10.85%

On average, 40 and 50 channels were chosen for 1. BCI Competition IV, Dataset 1, 2. BCI Competition III, Dataset IVA, respectively by cross validation method.

C. Classification Model

Many improvements to the CSP model have been made in recent years. And by improving the model they have tried to overcome the common problems of the traditional CSP method. Such methods are CSSP, RCSP, FBCSP, ERCSP etc. But in most cases, every method addresses one problem and there is always room for improvement. For example, the CSSP method was generated to extract Spatial spectral filters and thus improve the performance of the model. RCSP was invented to solve the problem caused for data deficiency in the training set. Our research's major goal is to address all of the problems at once and make a model which can overcome every issue with CSP. FBCSP (Filter Bank CSP) has done a major improvement by reducing the subject dependability. It separates the EEG frequency spectrum into multiple bands before using the CSP to extract features. In this research, we suggest a method for extracting features using several modified CSPs, followed by the use of Random Forest and SVM classifiers to categorize the data. Then we combine each prediction coming from the classifier after feeding a different set of features to it. It has been established that when the base learner's variance is high, the bagging strategy works effectively. Furthermore, the Random Forest classifier has a significant likelihood of overfitting. As the bagging tool, the Stacked Ensemble CSP and the Random Forest classifier are combined. The overfitting problem is mitigated in both models using the basic learner.

III. RESULTS AND DISCUSSION

The proposed approach is validated using two online publicly available BCI Competition datasets which have been discussed earlier. The suggested approach is compared to CSP and its modified versions since the purpose of this research is to improve the state-of-the-art technology based on the CSP algorithm. To make sure our method is performing well

we have conducted various types of experiments. First, we have compared the Stacked Ensemble CSP method with other versions of CSP methods that were proposed earlier. Next, we have evaluated our model's performance by feeding our data in Random Forest algorithm and compared the accuracy with other CSP methods like CSP, CSSP, RCSP, FBCSP algorithm. Then we have experimented to test our model with a random forest classifier and bagging technique. For this experiment, we have also applied the bagging technique with CSP+ RF, CSSP + RF, RCSP + RF, FBCSP + RF, and compared the result with the Stacked Ensemble CSP method. Then in the third phase, another experiment was conducted to check how our model is behaving with Random Forest classifier and SVM classifier with bagging technique. This experiment was also done with all versions of the CSP algorithm and compared the result with our model. A final experiment was carried out to assess one's performance of the preceding techniques with that of the suggested strategy. The performance of the offered technique was compared to that of a control group in the final experiment. earlier techniques that merely employed boosting to improve the classifier. The best accuracy in Tables 1, 2, 3, and 4 are marked in green color and the second-best results are marked in yellow color.

A. Experiment with BCI competition III Dataset IVA

There are two sections to this dataset: training and testing. The techniques were trained in the training section, and the methods were evaluated in the testing section. The major goal of the BCI Competition III Dataset IVA was to assess the competitors' algorithms' efficacy. The results of tests 1–4 using the Random Forest classifier are shown in Table 1. In Table 2, the SVM classifier was used to compare the outcomes of all of the CSP-based methods in Table 1. The block diagram in Fig. 1 was used to implement all of the techniques in Tables 1 and 2. It means that all methods used the same pre-processing stages. The proposed Stacked Ensemble CSP, which consists of CSP, CSSP, RCSP, FBCSP, the Random Forest classifier, and the bagging ensemble method, outperforms previous techniques for the BCI Competition III Dataset IVA dataset, as demonstrated in Tables 1 and 2.

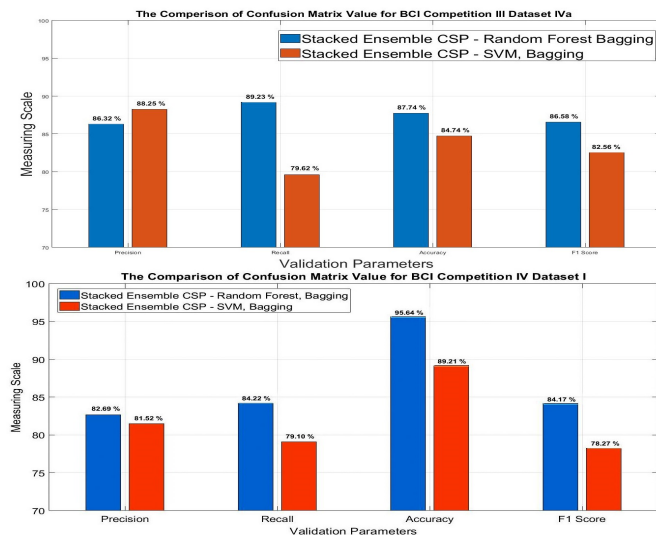


Fig. 3: Comparison of accuracy, recall, and precision for both datasets.

B. Experiment with BCI competition IV Dataset I

This dataset was collected including seven subjects denoted as a,b,c,d,e,f, and g. Each subject was gone through 200 trials and the EEG signals were collected for all of the trials. 5 fold cross-validation was applied to evaluate the above-mentioned methods. From Tables 3 and 4 it is almost clear that for this dataset our method's performance was insane compared to other advanced techniques. Along with bagging boosting is also giving promising results for this dataset (Table 4). The comparison of the confusion matrix value between two classifiers has been shown in Figure 3 for both of the datasets. And from the figure and table mentioned earlier, it can be concluded that RF+bagging with Stacked CSP features outperforms other methods.

IV. CONCLUSION AND FUTURE WORKS

In this work, a prototype for accurate brain to UAV communication has been proposed. An improved version of the CSP method has been proposed for improving the results for motor imagery EEG signal so that the real-time command passing to UAV through brain signal only can be possible. The various experiment was conducted to make sure that this method is best compared to the traditional approach. Also using the bagging technique the classifier performance has been enhanced. In summary, we have solved two significant CSP challenges in this work. Other flaws may occur as well; one issue is that CSP only considers one frequency band linked to the motor imagery task, although other bands may include useful information for diverse subjects. As a consequence, further research might overcome these difficulties, and an optimal CSP-based algorithm as a bagging technique base learner could be constructed.

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